

series reversal.

Next, to properly investigate the relationship between time series continuation and reversal, we perform both sign analysis and performance evaluation to determine the future returns of the above-mentioned portfolio segmentations. In the sign analysis, we find that the loser sub-portfolios, no matter whether their profits are realised or not, i.e. both realised losers and contrarian losers, experience strong subsequent reversal. By contrast, the results from the winner subgroups lead to mixed answers: the realised winners exhibit strong reversal, while the contrarian winners do not.

The results of performance evaluation again suggest that the loser subgroups, especially the contrarian losers, contribute the most to the time series reversal. This conclusion is consistent with the results about cross-sectional reversal noted by Conrad and Yavuz (2017); they do not observe statistically significant returns for realised portfolios (realised winners minus realised losers) but do observe significant returns for contrarian portfolios (contrarian winners minus contrarian losers). We take a step further by showing that the positive returns generated by the realised winner and the realised loser offset each other when considered together. However, the reversal of a contrarian loser is much stronger than that of a contrarian winner, which leads to more statistically significant results for contrarian portfolios.

A double-sorted trading strategy is then constructed, holding the four previously mentioned decompositions of trend-following strategies. We call this a “trend-following reversal strategy”, since it combines time series continuation and reversal. It differs from the widely documented cross-sectional reversal as it does not require any relative return comparisons. For each single asset, we first sort on the 12-month returns skipping the most recent 12 months (ranking period 1) and second sort on the recent 12-month returns (ranking period 2), and then invest in the following month. For instance, a realised winner strategy contains assets with positive returns in both ranking periods, 1 and 2. Figure 1 illustrates how these four double-sorted strategies are constructed. Our results suggest that holding the realised winner and the contrarian loser subgroups can yield annualised returns as high as 22% and 24%, respectively. These strategies are liquid and well diversified in the context of real market trading.